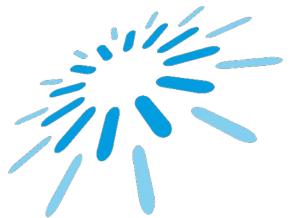




Collaboration, Quality, Test

Prof. Dr. Jóakim von Kistowski



TH Aschaffenburg
university of applied sciences

Recap:
Why can testing not show the **absence** of defects?

Chapter 1: Fundamentals of Testing

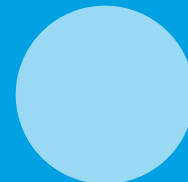
Terms and Motivation

Testing Principles

Test Activities, Testware and Test Roles

Test Cases, Expected Results, and Test Oracle

Essential Skills



Quiz

Question

**What phase in the testing process determines
“WHAT to test”?**

a) test planning

b) test analysis

c) test design

d) test implementation

Question

The test concept is created during the following phase:

a) test planning

b) test analysis

c) test design

d) test implementation

Question

Test design answers the question:

a) What is tested?

b) When to test?

c) Who tests?

d) How to test?

During test analysis, we ...

a) analyze the test basis
and check for ambiguities

b) prepare a test progress
report and draw up an
action plan

c) document the test
execution plan and select
test tools

d) document found errors

Question

A test object is ...

a) the work result to test

b) the source of the expected result
(→ test oracle)

c) The reason for testing
(→ test objective)

d) a number of test scripts
(→ test suite)

When to stop testing?

When the test objectives have been achieved!

NOT: When the time is up!

Unless we're out of money.

When to stop testing?



Based on coverage:

When everything I wanted to test was tested, e.g.:

- All features
- All quality attributes
- All platforms
- ...



Based on quality:

- Achieved sufficient quality level based on test results, e.g.: measured using number of errors per severity, etc.
- ...



Based on progress, e.g.:

- Finished all planned and necessary tests (according to test results)
- e.g.: Finished all error retests
- All areas that were re-prioritized due to test results

When to stop testing?



Based on coverage:

When everything I wanted to test was tested, e.g.:

- All features
- All quality attributes
- All platforms
- ...



Based on quality:

- Achieved sufficient quality level based on test results, e.g.: measured using number of errors per severity, etc.
- ...



Based on progress, e.g.:

- Finished all planned and necessary tests (according to test results)
- e.g.: Finished all error retests
- All areas that were re-prioritized due to test results

Assuming you are a project's test manager. What would be your professional reasons for not being willing to stop an ongoing test? Choose two criteria that might not have been met yet.

When to stop testing?



Based on coverage:

When everything I wanted to test was tested, e.g.:

- All features
- All quality attributes
- All platforms
- ...



Based on quality:

- Achieved sufficient quality level based on test results, e.g.: measured using number of errors per severity, etc.
- ...



Based on progress, e.g.:

- Finished all planned and necessary tests (according to test results)
- e.g.: Finished all error retests
- All areas that were re-prioritized due to test results

Assuming you are a project's test manager. What would be your professional reasons for accepting a software after system test?

Question

Why is test **analysis** important?

Why is test analysis important?

- Evaluation of test “input”
- Identifying known defects
(e.g. inconsistencies, inaccuracies, contradictions, etc.)
- Identification of feature set
- ...

Question

What can be used as **test basis**?

What can be used as test basis?

- Requirements specification, user stories, functional/non-functional requirements, use cases, personas ...
- Design documentation, e.g. UML diagrams, architecture descriptions, etc.
- Implementation of the component/system itself
- Risk analysis reports, error reports
- User documentation
- **ANYTHING** from which test cases can be derived!

Test Resources

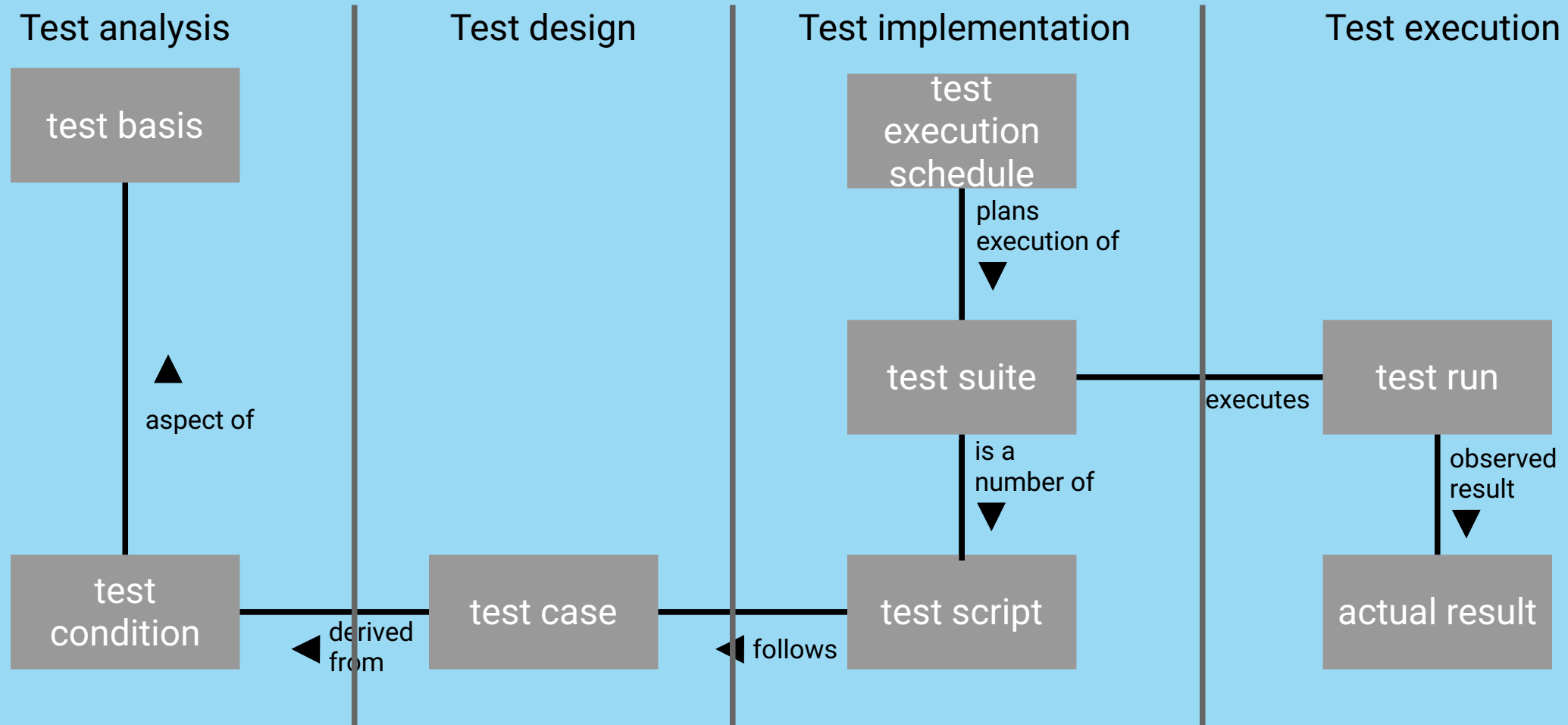
- All **artifacts created** during the testing process that are **required** to plan, design or execute tests.

Examples for **test resources**?

Test Resources

- All **artifacts created** during the testing process that are **required** to plan, design or execute tests.
- Includes: Documents, scripts, input data, expected results, procedures for setting up and cleaning up test data, files, databases, environments and other additional software and utilities used for testing.

Test Process Artifacts



Chapter 1: Fundamentals of Testing

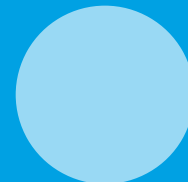
Terms and Motivation

Testing Principles

Test Activities, Testware and Test Roles

Test Cases, Expected Results, and Test Oracle

Essential Skills



Learning Objectives for Fundamentals of Testing

(according to Certified Tester Foundation Level Syllabus, English Version V4.0)

- **1.1 What is Testing?**
- FL-1.1.1 (K1) Identify typical test objectives
- FL-1.1.2 (K2) Differentiate testing from debugging
- **1.2 Why is Testing Necessary?**
- FL-1.2.1 (K2) Exemplify why testing is necessary
- FL-1.2.2 (K1) Recall the relation between testing and quality assurance
- FL-1.2.3 (K2) Distinguish between root cause, error, defect, and failure

Learning Objectives for Fundamentals of Testing

(according to Certified Tester Foundation Level Syllabus, English Version V4.0)

- **1.3 Testing Principles**

- FL-1.3.1 (K2) Explain the seven testing principles

- **1.4 Test Activities, Testware and Test Roles**

- FL-1.4.1 (K2) Summarize the different test activities and tasks

today FL-1.4.2 (K2) Explain the impact of context on the test process

- FL-1.4.3 (K2) Differentiate the testware that supports the test activities

today FL-1.4.4 (K2) Explain the value of maintaining traceability

- FL-1.4.5 (K2) Compare the different roles in testing

- **1.5 Essential Skills and Good Practices in Testing**

- FL-1.5.1 (K2) Give examples of the generic skills required for testing

- FL-1.5.2 (K1) Recall the advantages of the whole team approach

- FL-1.5.3 (K2) Distinguish the benefits and drawbacks of independence of testing

Test Process

- **test activities**
- **independent** process
- **closely interlinked** with software development
- Necessary: **fine granular execution plan** for tests in each **test stage**
- Specific software testing process depends on many factors
- **Test strategy** of a company:
 - Which test activities in the test process?
 - Use of the activities
 - When do which activities take place?

Question

**Name examples:
How does the test process depend on context?**

Test Process in Context (1/2)

Context-dependent factors influencing the test process:

- Software development life cycle model
 - → Process model, e.g., agile vs. classical
- Possible test levels and test types
 - → e.g., Do we need performance tests? Usability tests, etc.? Acceptance of third-party software vs. self-developed product
- Product and project risks
 - → e.g., Determine test intensity
- Business area
 - → in-house software vs. safety-critical component

Test Process in Context (2/2)

Context-dependent factors influencing the test process:

- Business restrictions
 - Budget and resources
 - Complexity
 - Deadlines, time limits
 - Contractual and regulatory requirements
- Guidelines and practices of the respective company
- Required internal and external standards

Learning Objectives for Fundamentals of Testing

(according to Certified Tester Foundation Level Syllabus, English Version V4.0)

- **1.3 Testing Principles**

- FL-1.3.1 (K2) Explain the seven testing principles

- **1.4 Test Activities, Testware and Test Roles**

- FL-1.4.1 (K2) Summarize the different test activities and tasks

today FL-1.4.2 (K2) Explain the impact of context on the test process

- FL-1.4.3 (K2) Differentiate the testware that supports the test activities

today FL-1.4.4 (K2) Explain the value of maintaining traceability

- FL-1.4.5 (K2) Compare the different roles in testing

- **1.5 Essential Skills and Good Practices in Testing**

- FL-1.5.1 (K2) Give examples of the generic skills required for testing

- FL-1.5.2 (K1) Recall the advantages of the whole team approach

- FL-1.5.3 (K2) Distinguish the benefits and drawbacks of independence of testing

Question

What does
“traceability between test basis and test work results”
mean?

What does “traceability between test basis and test work results” mean?

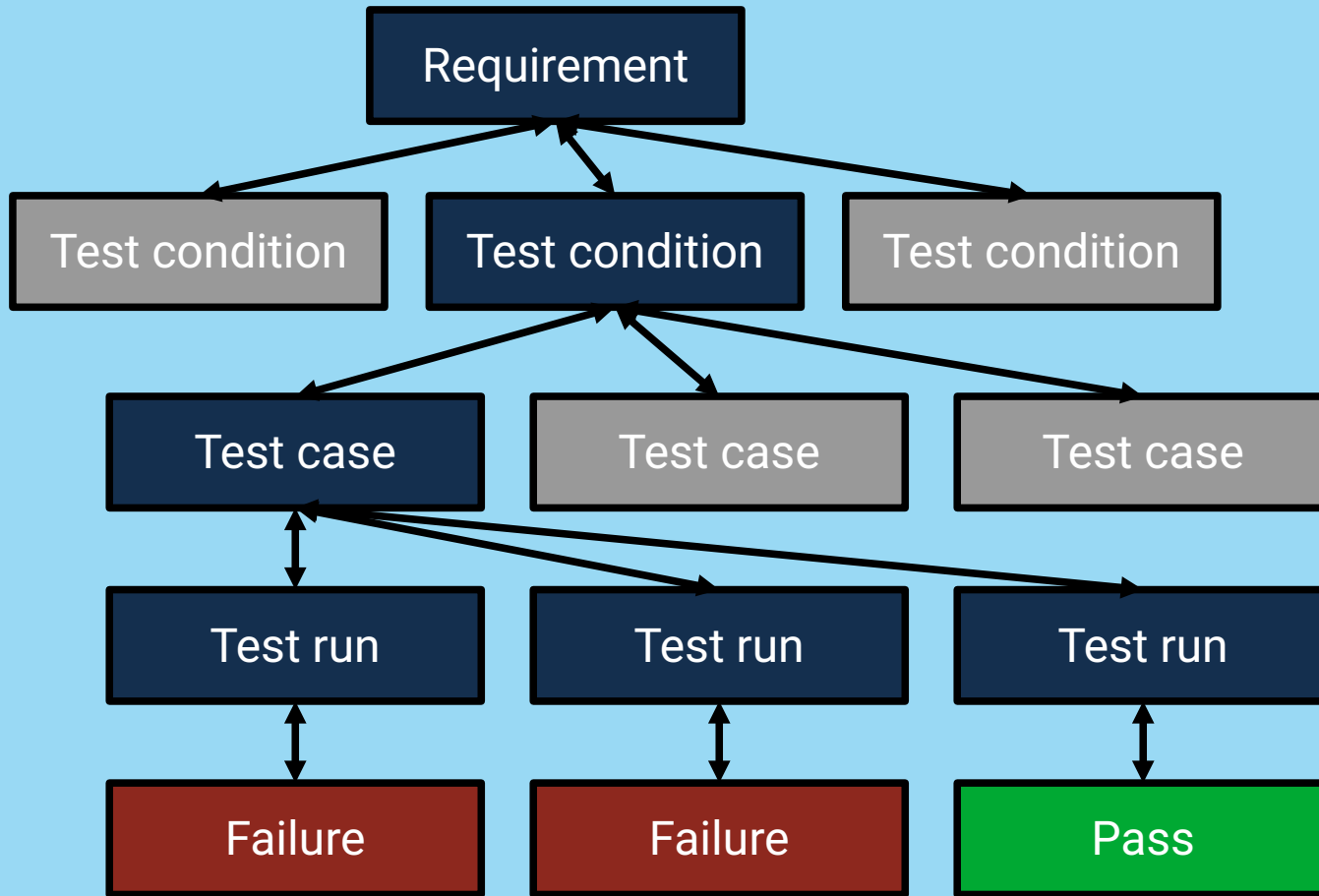
Establishing relationships between *each and every* element of the test basis and the results of the test activities

Traceability between test basis and test work results

Traceability:

- The degree to which a relationship can be established between two or more products of the development process [...] [ISO 19506]

Traceability



We want to answer the following:

- Is each requirement covered by at least x test cases?
- Is each test condition covered by at least x test cases?
- What percentage of the test cases were executed?
- Which test cases have not yet been executed?
- How many defects were detected per requirement?
- What percentage of the test cases show failures?
- ...

Why is traceability important?

**Test control is not possible without
bidirectional traceability!**

Traceability between test basis and test work results (1/3)

Traceability benefits: evaluation of test coverage

- Change impact analysis: Which test cases are affected by a change → Consequences on test effort
- Traceability of tests
- Improve the comprehensibility of test progress reports and test completion reports
 - Not exclusively qualitative statements, instead:
80% of test cases were executed, 20% show failures, ...
- Fulfill IT governance criteria → Specifications e.g. regarding coverage
- Provide information to assess product quality, process capability and project progress against business objectives

Challenges

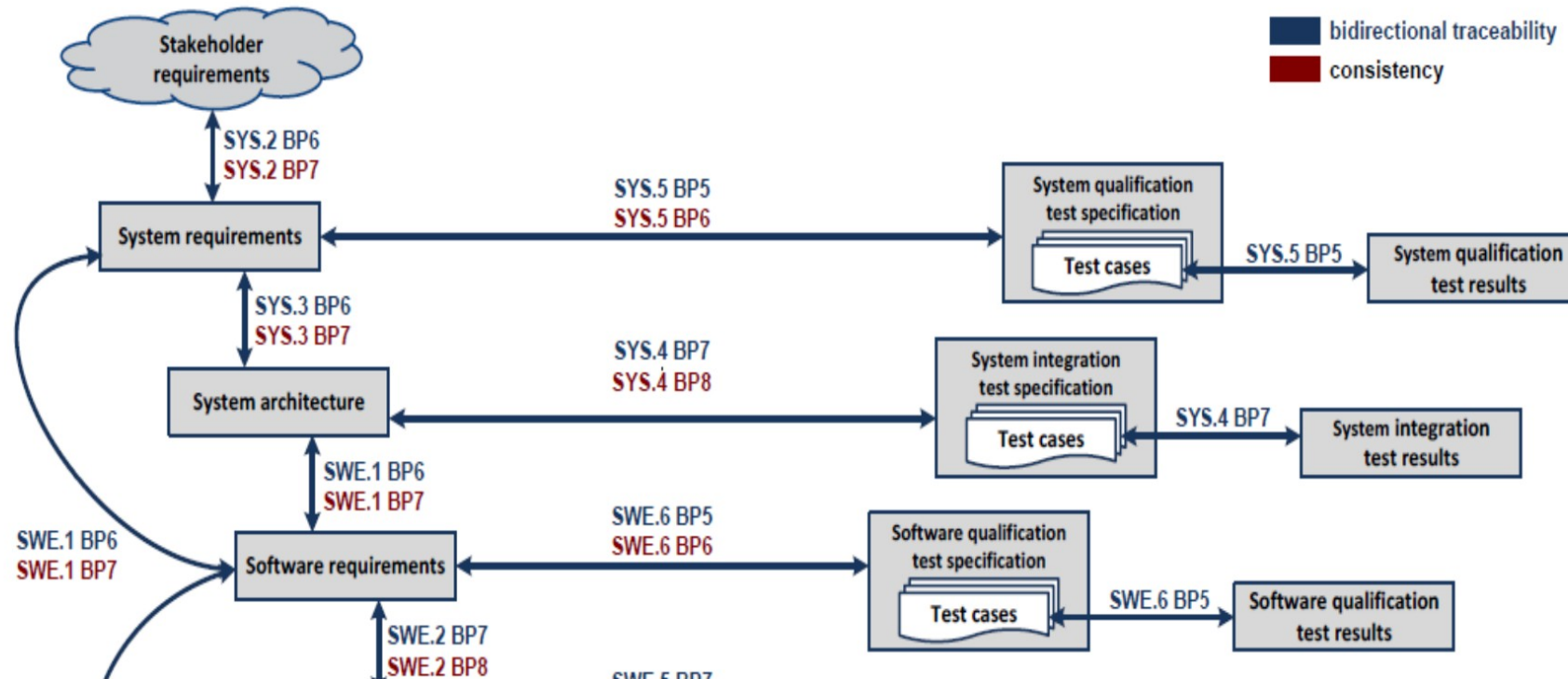
- Partly manual → incomplete, error-prone
- Main beneficiaries may be other people (no “own pain”, “only” effort)

Traceability between test basis and test work results (2/3)

Furthermore, bidirectional traceability has been explicitly defined between

- test cases and test results, and
- change requests and work products affected by these change requests.

An overview of bidirectional traceability and consistency is depicted in the following figure.



Source: Automotive SPICE 3.1

Traceability between test basis and test work results (3/3)

Traceability

- The degree to which a relationship can be established between two or more products of the development process [...] [ISO 19506]

Vertical Traceability

- Tracing of requirements from development documentation to components.

Horizontal traceability

- Tracing requirements of a test level through the levels of test documentation (e.g., test concept, test design specification, test case specification, test procedure specification or test scripts).

Chapter 1: Fundamentals of Testing

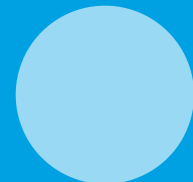
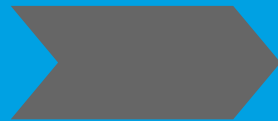
Terms and Motivation

Testing Principles

Test Activities, Testware and Test Roles

Test Cases, Expected Results, and Test Oracle

Psychology of Testing



A hand with long, pointed fingers holds a large, glowing golden sphere. The sphere has a textured, metallic surface with some darker, scuffed areas. It emits a bright, warm light from its center, creating a lens flare effect. The background is a solid, vibrant blue.

Test oracle

Expected Results and Test Oracle

- After each test case execution: Failure yes / no?
 - Compare actual result with expected result
- Expected result of the test object must be specified in advance
 - Task of tester: find suitable sources
- “Test oracle” is consulted to predict expected results

Test Oracle

- Expected results must be derived from specification.
- Three possibilities:
 - **Tester derives the expected result**
Basis: Test object spec & input data (**usual way of doing things**)
 - **Formal specification available?**
Create executable prototype with tool support
Results of the prototype serve as a test oracle for the actual test
 - **Back-to-back test**
 - software created multiple times by independent development groups
 - results of all program versions are compared (with the same inputs)
Different results: Defect in one version
 - Only defects undetected that have the same error effects in all versions

Further Test Oracles

- Use user manual
- Previous versions (tested!), programs with similar functionality
- Exact values cannot always be predicted or determined
→ Define tolerance range, check plausibility
- Experience is important
- System itself (create usage profile) -> worst option

Chapter 1: Fundamentals of Testing

Terms and Motivation

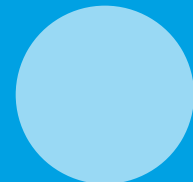
Testing Principles

Test Activities, Testware and Test Roles

Test Cases, Expected Results, and Test Oracle



Essential Skills



How do testers have to think differently from developers?

Traceability between test basis and test work results

Focus
Development

Ways of Thinking Development / Test

Focus
Test

- Design solutions
- Confirm solutions
→ Warning: Confirmation Bias

- Test and validate solutions
- Critically scrutinize the solution
- Curiosity and attention to detail
- High level of positive communication and empathy

Traceability between test basis and test work results

Focus
Development

Ways of Thinking Development / Test

Focus
Test

- Design solutions
- Confirm solutions
→ Warning: Confirmation Bias

→ successful projects combine
both ways of thinking

- Test and validate solutions
- Critically scrutinize the solution
- Curiosity and attention to detail
- High level of positive communication and empathy



Take Away – Chapter 1 – Fundamentals of Testing (1/4)

Failure - Non-fulfillment of a requirement!

- Description of such a situation as failure
- Cause: Defect in the software
- caused by person's error

Tests are important measures for ensuring the **quality** attributes

- Functionality, reliability, usability, IT security, compatibility, portability, maintainability and performance efficiency [according to ISO 25010]



Take Away – Chapter 1 – Fundamentals of Testing (2/4)

- Testing everything is impossible, tests are always **random samples**,
→ failures can only be proven with a certain probability
- Intensity and scope of the tests depend on the expected risk in the event of incorrect program behavior
- **Prioritize tests**



Take Away – Chapter 1 – Fundamentals of Testing (3/4)

The test process consists of the following activities

- test planning
- test monitoring and control
- test analysis
- test design
- test implementation
- test execution
- test completion

Test case: input value; expected result; preconditions under which the test conditions under which the test case must run, and post-conditions that must be fulfilled after execution.



Take Away – Chapter 1 – Fundamentals of Testing (4/4)

- Test case execution: Test object shows actual result
 - Are expected and actual results different?
 - Check whether expected result, test case, specification, or test object, contain errors
- A test oracle determines the target values for each test case before test execution.



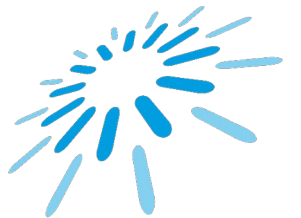
You should now be able to answer these questions

- Define the terms: Error, Fault, Failure
- Explain the difference between testing and debugging.
- Define the terms verification and validation.
- Explain why every test is a random sample test.
- Name the main characteristics of software quality according to ISO.
- Define the term reliability of a system.
- Explain the main activities of the testing process.
- Characterize the work results associated with the steps of the test process.
- Explain bi-directional traceability.
- What is a test oracle?



Until next time!

Collaboration, Quality, Test
Prof. Dr. Jóakim von Kistowski



TH Aschaffenburg
university of applied sciences